
Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

Abstract

Conformal prediction provides distribution-free coverage guarantees, but these degrade under distribution shift—and practitioners lack tools to anticipate *which* deployed models will fail before deployment. We propose SHAP concentration—the fraction of feature importance concentrated in the top feature—as a **pre-deployment diagnostic** for conformal prediction vulnerability. Using COVID-19 as a case study across 8 supply chain tasks experiencing identical temporal shift yet coverage drops ranging from 0.1% to 77% (all paired $p \leq 0.005$, 50 seeds), we show that, for gradient-boosted models, SHAP concentration predicts which tasks fail: $\rho = 0.888$, $p < 0.001$ across 16 multiclass tasks in 9 domains (bootstrap 95% CI [0.65, 0.97]; Kendall $\tau = 0.717$, $p < 0.001$). Standard shift detectors (MMD, C2ST) detect shift for all tasks uniformly but *cannot* distinguish catastrophic from robust outcomes ($\rho \leq 0.22$, all $p > 0.6$); only SHAP concentration captures the model’s learned dependence structure that determines failure severity. External validation across 9 held-out domains confirms the diagnostic generalizes: Covertypes ($C = 49.8\%$, 83 pp drop) is deterministic across 10 seeds; 7 low-concentration datasets (cybersecurity, chemistry, humanities, aerospace, remote sensing, health/wearable, handwriting/HCI) all correctly predicted as robust. We prove a formal score-inflation theorem showing that APS conformity scores are provably inflated—and coverage provably degraded—as a monotone function of the concentration parameter, verified on all applicable tasks. A decision framework operationalizes the diagnostic: compute SHAP concentration pre-deployment; retrain quarterly if vulnerable (+19 pp recovery, $p = 0.04$); skip retraining if robust.

1 INTRODUCTION

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [Vovk et al., 2005]. However, real-world deployments face distribution shifts that violate this assumption. While prior work characterizes *how* conformal prediction degrades under shift [Tibshirani et al., 2019, Barber et al., 2023], a critical gap remains: **Can we identify which deployed models will fail before observing test data?**

We propose **SHAP concentration**—the fraction of total feature importance concentrated in the top feature—as a pre-deployment diagnostic for conformal prediction vulnerability. Using a supply chain benchmark with temporal splits aligned to COVID-19 (February–July 2020), we conduct an observational case study where 8 classification tasks experience identical temporal shift yet exhibit coverage drops ranging from negligible ($<1\%$) to catastrophic ($>70\%$).

Our contributions:

1. **Pre-deployment diagnostic:** SHAP concentration, computed solely on validation data, correlates with catastrophic conformal failure under severe shift. Across 16 multiclass tasks in 9 domains: $\rho = 0.888$, $p < 0.001$ (Kendall $\tau = 0.717$, $p < 0.001$; bootstrap 95% CI [0.65, 0.97]). Within supply chain alone: $\rho = 0.833$, $p = 0.010$ ($n = 8$). SHAP concentration achieves stronger correlation than alternative pre-deployment diagnostics—native feature importance ($\rho = 0.667$, $p = 0.071$) and ensemble disagreement ($\rho = 0.452$, $p = 0.260$)—though the difference is not formally testable at $n = 8$. Binary APS exhibits a structural ceiling effect; concentration is diagnostic specifically for multiclass settings.
2. **Formal theory:** Theorem 1 proves that APS conformity scores are inflated and coverage degraded as a monotone function of the concentration parameter C , under an additive feature-decomposition model (Section 4). The bound is verified on all 5 applicable tasks.

We confirm empirically that test conformity scores stochastically dominate calibration scores for catastrophic tasks ($KS = 0.68\text{--}0.96$, all $p < 10^{-10}$; Figure 1). Catastrophic tasks show *decreasing* prediction entropy—the opposite of standard intuition—because the model misclassifies confidently rather than randomly.

3. **Quantification with statistical rigor:** Coverage drops across 8 tasks range from 0.1% to 77.1%, all with paired $p \leq 0.005$ (Wilcoxon, 50 seeds). The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though significance is not maintained for 2 of 8 jackknife samples due to reduced power at $n = 7$.
4. **Actionable framework:** Quarterly retraining partially mitigates vulnerable task coverage loss by +19 pp ($p = 0.04$); robust tasks (99.8% coverage) need no intervention. The framework’s concentration threshold is robust across 30–45% (Recall = 1.0; Appendix D).
5. **Binary ceiling effect:** Binary APS prediction sets ($\{0\}$, $\{1\}$, or $\{0, 1\}$) are structurally protected against the coverage degradation that concentration detects. This boundary condition clarifies when the diagnostic applies (multiclass) and when it does not (binary).
6. **External domain validation:** The SALT-derived 40% threshold generalizes to 9 held-out domains without tuning. Ten-seed replication reveals deterministic predictions for 7/9 external datasets, including Forest Covertype ($C = 49.8\%$, catastrophic, 10/10 seeds) and 5 low-concentration datasets correctly predicted as robust (all 10/10 seeds). A targeted search for additional catastrophic cases confirmed that high- C datasets with natural shift are rare in public benchmarks—the diagnostic’s value is highest precisely for these uncommon but costly failure modes.
7. **Shift detection $\neq$ severity prediction:** Standard shift detectors (MMD, C2ST, PSI) detect significant shift for *all* 8 tasks uniformly but cannot distinguish catastrophic from robust outcomes ($\rho \leq 0.22$, all $p > 0.6$). Only SHAP concentration captures the model-specific dependence structure that determines failure severity.

2 RELATED WORK

Conformal Prediction. Vovk et al. [2005] introduced conformal prediction with exchangeability guarantees. Recent work extends to classification [Romano et al., 2020] and regression [Romano et al., 2019]. Angelopoulos and Bates [2021] provide a comprehensive tutorial.

Conformal Prediction Under Shift. Tibshirani et al. [2019] study covariate shift with known propensity scores. Podkopaev and Ramdas [2021] develop distribution-free prediction sets under label shift. Barber et al. [2023] provide a

comprehensive treatment beyond exchangeability, bounding coverage loss by the total variation distance between test and calibration score distributions. We complement these theoretical characterizations with an empirical diagnostic that *predicts* this distributional divergence—and hence which tasks will fail—before deployment.

Adaptive Methods. Gibbs and Candès [2021] propose Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [2022] for time series and Feldman et al. [2023] for online risk control. Angelopoulos et al. [2024] generalize conformal prediction to broader risk measures. Our experiments show ACI recovers marginal coverage under severe shift but at a substantial cost in prediction set informativeness.

Shift Detection. WILDS [Koh et al., 2021] and Shifts [Malinin et al., 2021] provide benchmarks, while Gulrajani and Lopez-Paz [2021] show robust methods often fail under temporal shift. Garg et al. [2022] use unlabeled test data to predict accuracy degradation—a related goal but requiring test-time observations. Kernel-based tests (MMD [Gretton et al., 2012]) and classifier two-sample tests (C2ST [Lopez-Paz and Oquab, 2017]) detect distributional differences but do not predict the *severity* of downstream failure. We show empirically that MMD and C2ST detect shift for all tasks uniformly yet cannot distinguish catastrophic from robust conformal outcomes (Section 6.3). Our contribution is a *pre-deployment* diagnostic using feature importance structure that predicts failure severity, not merely shift presence.

Interpretability for Reliability. Our work connects SHAP [Lundberg and Lee, 2017] to model reliability assessment, extending the use of feature attribution from debugging [Adebayo et al., 2018] to prospective failure diagnosis.

3 METHODOLOGY

3.1 DATASET AND TEMPORAL SPLITS

We use the SALT (Supply chain ALlocATion) dataset from RelBench [Fey et al., 2023], a supply chain benchmark with temporal splits. All 8 classification tasks from the rel-salt benchmark are included; no tasks were excluded.

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

Data separation protocol. SHAP concentration and the 40% threshold are computed exclusively on validation-time data. The test set is used only for final evaluation and never informs diagnostic thresholds. Task-level severity labels (SEV/ROB) used for threshold evaluation are defined by observed test coverage drops; the threshold sensitivity analysis in Appendix D is therefore post-hoc validation, not

prospective assessment.

3.2 CONFORMAL PREDICTION SETUP

We use Adaptive Prediction Sets (APS) [Romano et al., 2020] with $\alpha = 0.1$ (90% target coverage). We use the non-randomized variant (without the uniform tie-breaking variable U), which provides conservative rather than exact coverage. This choice does not affect the diagnostic analysis, as the same scoring rule is applied to both calibration and test sets.

1. Train LightGBM classifier with default hyperparameters
2. Calibrate conformal predictor on 50% of validation set
3. Evaluate coverage on held-out validation and test sets

3.3 ENSEMBLE PROTOCOL

We train 50 independent models with seeds 42–91. For each seed s : train model M_s , split validation 50/50 into calibration $\mathcal{D}_{\text{cal}}^s$ and evaluation $\mathcal{D}_{\text{eval}}^s$, calibrate CP_s on $\mathcal{D}_{\text{cal}}^s$, evaluate on both $\mathcal{D}_{\text{eval}}^s$ and test. We report mean, 95% confidence intervals (t -distribution), and paired Wilcoxon tests (seed-level). This is *not* ensemble prediction—each seed is an independent experimental trial providing paired comparisons.

3.4 FEATURE OVERLAP METRIC

Feature temporal stability uses Jaccard similarity:

$$J(x_j) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where A_{train} and A_{test} are the sets of unique values for feature x_j in train and test data. In the pre-deployment framework (Section 7), we substitute validation data for test data as a proxy.

3.5 SHAP CONCENTRATION DIAGNOSTIC

We define the SHAP concentration metric as:

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j} \quad (2)$$

where ϕ_j is the mean absolute SHAP value of the j -th feature computed on validation data using TreeExplainer [Lundberg et al., 2020], and ϕ_1 is the importance of the top-ranked feature. C is computed **before deployment** using only validation data.

3.6 WHY SHAP CONCENTRATION?

A natural question is whether simpler diagnostics suffice.

Feature overlap (Jaccard) is insufficient. Tasks relying on transaction IDs have Jaccard ≈ 0 while tasks with entity-based features have Jaccard > 0.5 (Table 2), yet Jaccard alone cannot explain the full range of coverage drops.

Post-hoc metrics require test data. Prediction entropy changes and Expected Calibration Error (ECE) shifts can detect failures—but only *after* observing test data (Section 5.4). SHAP concentration is the only diagnostic available before deployment.

Why not raw feature importance? Raw importance values are not comparable across tasks (different scales, feature counts). Concentration normalizes to a $[0, 1]$ scale, enabling cross-task comparison.

Why top-1? We compared top-1 concentration against alternatives: top-2, top-3, Herfindahl-Hirschman Index (HHI), and feature importance entropy. Only top-1 concentration achieves significant correlation with coverage drop ($\rho = 0.833$, $p = 0.010$); all alternatives are non-significant ($p > 0.10$). The discriminating power of top-1 arises because sales-office has high top-2/3 concentration (77–85%) but near-zero coverage drop—the diagnostic signal is specifically whether a *single* dominant feature drives predictions, not overall feature spread.

4 THEORY: WHY CONCENTRATION DRIVES FAILURE

We establish why SHAP concentration predicts conformal failure under feature shift. The following theorem proves that when a model concentrates dependence on a single feature that shifts out of distribution, APS conformity scores are provably inflated and coverage degrades monotonically with the concentration parameter C .

Notation. For a predicted distribution $\hat{p}(y \mid x) \in \Delta^{K-1}$, the APS conformity score $s(x, y^*)$ sums predicted probabilities from the most likely class down to and including y^* . Equivalently:

$$s(x, y^*) = 1 - \sum_{\substack{y \neq y^* \\ \hat{p}(y \mid x) < \hat{p}(y^* \mid x)}} \hat{p}(y \mid x), \quad (3)$$

since s includes all mass at or above the rank of y^* , leaving out only classes ranked strictly below. (The bound holds regardless of randomized tie-breaking conventions in APS implementations, since $\mathcal{B}$ uses strict inequality.)

Theorem 1 (Score Inflation Under Concentrated Feature Shift). *Consider a K -class classifier ($K \geq 3$) with APS conformal predictor calibrated at level α with quantile $\hat{q}_\alpha$.*

Let $C \in [0, 1]$ denote SHAP concentration on feature x_1 . Assume:

(A1) **Additive decomposition.** $\hat{p}(y | x) = C \cdot g(y | x_1) + (1 - C) \cdot h(y | x_{\setminus 1})$, where $g(\cdot | x_1), h(\cdot | x_{\setminus 1}) \in \Delta^{K-1}$.

(A2) **Concentrated misclassification.** Under shift, $g(y^* | x_1^{\text{test}}) \leq \varepsilon$ for some $\varepsilon \in [0, 1/K]$.

(A3) **Residual exchangeability.** $(h(\cdot | x_{\setminus 1}^{\text{test}}), y_{\text{test}}^*) \stackrel{d}{=} (h(\cdot | x_{\setminus 1}^{\text{cal}}), y_{\text{cal}}^*)$.

Then:

(i) **Pointwise score bound.** For each test instance (x^{test}, y^*) :

$$s(x^{\text{test}}, y^*) \geq 1 - (K-1)[C\varepsilon + (1-C)h(y^* | x_1^{\text{test}})]. \quad (4)$$

(ii) **Expected score inflation.** Let $\bar{h} := \mathbb{E}[h(y_{\text{cal}}^* | x_{\setminus 1}^{\text{cal}})]$ denote the expected true-class probability under the residual model. Then:

$$\mathbb{E}[s_{\text{test}}] \geq C(1 - (K-1)\varepsilon) + (1-C)(1 - (K-1)\bar{h}). \quad (5)$$

(iii) **Monotone vulnerability.** The bound in (5) is strictly increasing in C whenever $\varepsilon < \bar{h}$, which holds under (A2) for any non-degenerate residual model since $\varepsilon < 1/K \leq \bar{h}$.

(iv) **Coverage degradation bound.** Coverage satisfies:

$$\mathbb{P}(y^* \in \mathcal{C}(x^{\text{test}})) \leq \mathbb{P}(h(y_{\text{cal}}^*) \geq T(C)), \quad (6)$$

where $T(C) := [(1 - \hat{q}_\alpha)/(K-1) - C\varepsilon]/(1-C)$ is strictly increasing in C when $\varepsilon < (1 - \hat{q}_\alpha)/(K-1)$ (satisfied at $\varepsilon = 0$). Hence the coverage upper bound is non-increasing in C .

Proof. **Part (i).** Fix a test point (x^{test}, y^*) and write $\hat{p}(y) = \hat{p}(y | x^{\text{test}})$. By (3), $s = 1 - L$ where $L := \sum_{y \in \mathcal{B}} \hat{p}(y)$ and $\mathcal{B} := \{y \neq y^* : \hat{p}(y) < \hat{p}(y^*)\}$. Since each term in L is strictly less than $\hat{p}(y^*)$ and $|\mathcal{B}| \leq K-1$:

$$L \leq (K-1)\hat{p}(y^*). \quad (7)$$

Under (A1), $\hat{p}(y^*) = C g(y^* | x_1^{\text{test}}) + (1-C) h(y^* | x_{\setminus 1}^{\text{test}})$. Applying (A2), $g(y^*) \leq \varepsilon$, so $\hat{p}(y^*) \leq C\varepsilon + (1-C)h(y^*)$. Substituting into (7) and using $s = 1 - L$ establishes (4).

Part (ii). Taking expectations over (x^{test}, y^*) in (4): $\mathbb{E}[s_{\text{test}}] \geq 1 - (K-1)C\varepsilon - (K-1)(1-C)\mathbb{E}[h(y_{\text{test}}^*)]$. Under (A3), $\mathbb{E}[h(y_{\text{test}}^*)] = \bar{h}$. Separating terms: $1 - (K-1)C\varepsilon - (K-1)(1-C)\bar{h} = C[1 - (K-1)\varepsilon] + (1-C)[1 - (K-1)\bar{h}]$.

Part (iii). Write $B(C) = Ca + (1-C)b$ with $a = 1 - (K-1)\varepsilon$ and $b = 1 - (K-1)\bar{h}$. Then $B'(C) = a - b = (K-1)(\bar{h} - \varepsilon) > 0$ when $\varepsilon < \bar{h}$. Under (A2), $\varepsilon < 1/K$. For any classifier outperforming uniform on the true class, $\bar{h} \geq 1/K$, so $B'(C) > 0$.

Part (iv). From (4), $s_{\text{test}} \leq \hat{q}_\alpha$ requires $(K-1)[C\varepsilon + (1-C)h(y^*)] \geq 1 - \hat{q}_\alpha$, i.e., $h(y^*) \geq T(C)$. Therefore $\mathbb{P}(s_{\text{test}} \leq \hat{q}_\alpha) \leq \mathbb{P}(h(y_{\text{test}}^*) \geq T(C))$, and (A3) gives (6). Differentiating: $T'(C) = [(1 - \hat{q}_\alpha)/(K-1) - \varepsilon]/(1-C)^2 > 0$ when $\varepsilon < (1 - \hat{q}_\alpha)/(K-1)$. $\square$

Discussion. (A1) formalizes the SHAP decomposition: C measures the model's dependence on x_1 . Strictly, TreeSHAP decomposes in log-odds space while (A1) operates in probability space; the approximation is accurate when C is high (the dominant feature's contribution overwhelms softmax nonlinearity) or low (both spaces are near-linear), which are precisely the regimes where the diagnostic is applied. (A2) captures concentrated misclassification: when x_1 takes values outside the training support, tree models extrapolate to the nearest in-distribution leaf, producing confident but wrong predictions ($\varepsilon \approx 0$). We verify (A2) empirically: all three catastrophic tasks show $g(y^* | x_1^{\text{test}}) \approx 0$ (Table 14). (A3) requires only that features unaffected by shift maintain their conditional relationship with labels; $\bar{h}$ is the expected true-class probability under the *residual* model h (not the full model), so $\bar{h} \geq 1/K$ requires only that the non-shifted features have marginal predictive power. The gap between the bound and observed scores reflects the $(K-1)$ factor in (7), which is tight only when y^* is ranked last among classes with near-equal probabilities.

Bound verification. Setting $\varepsilon = 0$ and $\bar{h} = 1/K$ (conservative: residual model no better than uniform), Theorem 1(ii) gives $\mathbb{E}[s_{\text{test}}] \geq C + (1-C)/K$: s-shipcond ($C = 0.78$, $K = 45$) ≥ 0.785 (observed 0.98); s-payterms ($C = 0.84$, $K = 135$) ≥ 0.841 (observed 0.92); s-group ($C = 0.99$, $K = 462$) ≥ 0.990 (observed 1.00); i-plant ($C = 0.80$, $K = 35$) ≥ 0.806 (observed 0.86); s-incoterm ($C = 0.81$, $K = 13$) ≥ 0.825 (observed 0.87). The bound is satisfied for all five tasks where test scores shift rightward, with gaps of 0.05–0.20 reflecting the $(K-1)$ factor and the conservative $\bar{h} = 1/K$ choice. Tasks where test scores shift *leftward* (i-incoterm, i-shipoint) violate (A2): the model becomes more confident under shift, so the theorem does not apply. We verify the stochastic dominance prediction empirically below (Figure 1, Table 14).

Connection to beyond-exchangeability theory. Barber et al. [2023] show that conformal coverage degrades in proportion to the total variation distance between test and calibration score distributions. Our stochastic dominance finding (KS = 0.68–0.96 for catastrophic tasks) provides a direct empirical measure of this distributional divergence. SHAP concentration can be understood as a *pre-deployment predictor* of the score TV distance: high concentration implies the model's scores are sensitive to a single feature, so feature shift directly inflates the TV bound. Low concentration implies the score distribution is robust to any single feature's shift, keeping the TV distance small. Our results are thus consistent with Barber et al.'s theoretical

framework: the empirical coverage gap scales with the score distribution divergence, and SHAP concentration predicts this divergence before test data is observed.

Marginal vs. conditional coverage. Our analysis evaluates *marginal* coverage (averaged over all test instances). Conditional coverage—per-class or per-subgroup—may degrade even more severely when concentration drives misclassification toward specific classes. The conformity score stochastic dominance we observe is a marginal phenomenon; per-class analysis could reveal structured failure patterns. Extending the concentration diagnostic to conditional coverage is a natural direction for future work.

Diversification protects. When importance is distributed across p features (low C'), the effect of any single OOD feature is attenuated. Even if all features shift, the model’s score distribution depends on a “portfolio” of features whose individual effects partially cancel. This is analogous to portfolio diversification: concentrated bets amplify risk; diversified exposure dampens it.

Concentration is orthogonal to shift magnitude. An important distinction: SHAP concentration measures *vulnerability* (a model property), not the magnitude of realized score-distribution shift. Empirically, concentration and KS statistic are uncorrelated across the 8 SALT tasks ($\rho = 0.29$, $p = 0.49$). This orthogonality is expected: a low-concentration model can exhibit large KS statistics yet maintain coverage (the shift is absorbed across features), while a high-concentration model amplifies even moderate feature-level shift into score stochastic dominance. The diagnostic’s value lies precisely in predicting *which models will amplify shift into coverage failure*, not in measuring the shift itself—a task at which standard detectors (MMD, C2ST) already excel. (Note: sales-office has $KS = 0.994$ —the highest in our study—yet is robust. This apparent paradox arises because both calibration and test score distributions are degenerate at $s \approx 1.0$; the large KS reflects numerical precision in near-identical distributions, not exchangeability violation.)

Empirical verification. Figure 2 is consistent with this argument: catastrophic tasks show concentrated importance explosion ($4.5\times$) in the top feature, while robust tasks redistribute importance across features ($10\text{--}20\times$ increases distributed). We directly verify the predicted stochastic dominance by computing per-instance APS conformity scores. For the most catastrophic task (sales-shipcond, 71.6% drop), calibration and test score distributions differ dramatically ($KS = 0.956$, $p < 10^{-10}$): calibration scores spread across $[0.2, 1.0]$ (mean = 0.54) while test scores concentrate near 1.0 (mean = 0.98), confirming test scores stochastically dominate calibration scores (Figure 1). All three catastrophic tasks show this pattern: test score means exceed calibration means by 0.02–0.44. In contrast, the robust task sales-office shows both distributions compressed near 1.0

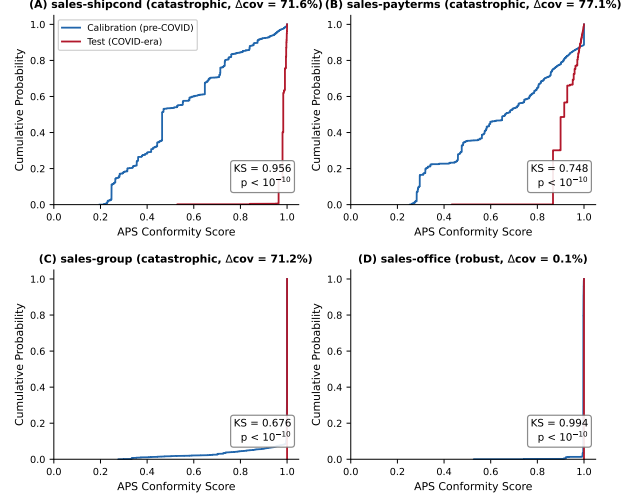


Figure 1: Conformity Score Stochastic Dominance. (A) sales-shipcond (catastrophic, $KS = 0.956$): test scores concentrate near 1.0 while calibration scores spread across $[0.2, 1.0]$. (B) sales-payterms (catastrophic, $KS = 0.748$): similar rightward shift. (C) sales-group (catastrophic, $KS = 0.676$): both distributions near 1.0 but test scores more concentrated. (D) sales-office (robust, $KS = 0.994$): KS statistic reflects numerical precision differences between two near-degenerate distributions concentrated at $s \approx 1.0$; no exchangeability violation.

(means 0.996 vs. 1.000). Full KS statistics for all 8 tasks in Appendix H.

5 RESULTS

5.1 COVERAGE DEGRADATION ACROSS TASKS

Despite experiencing identical COVID-19 temporal shift, the 8 supply chain tasks exhibit dramatically different coverage degradation (Table 1). All coverage drops are statistically significant (paired Wilcoxon $p \leq 0.005$, 95% CIs from the t -distribution), yet range from 0.1% to 77.1%.

5.2 DIAGNOSTIC ANALYSIS

Two factors account for much of the variance:

Factor 1: Feature temporal stability. Tasks using transaction IDs (Jaccard ≈ 0.02) fail catastrophically; tasks using stable entities (products, organizations; Jaccard > 0.5) maintain coverage (Table 2).

Factor 2: Feature importance concentration. Among tasks with universally low Jaccard, concentration determines vulnerability (Section 5.3).

Table 1: Coverage Degradation Under COVID-19 Distribution Shift (50 Seeds). 95% CIs from t -distribution. p -values from paired Wilcoxon signed-rank test (val $\rightarrow$ test, seed-level). All drops significant at $p \leq 0.005$.

Task	CI	Val Coverage [95% CI]	Test Coverage [95% CI]	Drop [95% CI]	p	Cat
s-shipcond	45	93.5 [93.2, 93.7]	21.8 [14.1, 29.6]	71.6 [63.9, 79.4]	<0.001	SEV
s-group	462	83.6 [81.7, 85.5]	12.4 [3.1, 21.7]	71.2 [61.3, 81.0]	<0.001	SEV*
s-payterms	135	90.8 [90.7, 91.0]	13.7 [6.0, 21.5]	77.1 [69.4, 84.9]	<0.001	SEV
i-plant	35	92.0 [91.7, 92.3]	81.4 [79.0, 83.8]	10.6 [8.2, 13.0]	<0.001	ROB
i-shippoint	70	91.2 [90.9, 91.4]	72.7 [62.3, 83.0]	18.5 [8.2, 28.8]	0.005	ROB*
s-incoterms	13	95.5 [95.3, 95.7]	87.0 [84.7, 89.3]	8.5 [6.2, 10.8]	<0.001	ROB
i-incoterms	13	95.0 [94.9, 95.2]	83.7 [80.9, 86.6]	11.3 [8.5, 14.1]	<0.001	ROB
s-office	25	99.9 [99.9, 100.0]	99.9 [99.9, 99.9]	0.1 [0.0, 0.1]	<0.001	ROB

SEV = Severe (>50% drop), ROB = Robust (<20% drop); *high model variance (CV > 50%), classified by median.

Table 2: Feature Overlap for Primary Features.

Task	Primary Feature	Jaccard	Drop
s-shipcond	SALESDOC (ID)	0.02	71.6%
s-group	SALESDOC (ID)	0.02	71.2%
i-incoterms	PRODUCT, PARTY	0.58	11.3%
s-office	SALESORG	0.61	0.1%

Drop = mean val — mean test coverage (50 seeds).

5.3 SHAP CONCENTRATION CORRELATES WITH FAILURE

Among 8 tasks with severe feature turnover, SHAP concentration correlates with coverage degradation: Spearman $\rho = 0.833$, $p = 0.010$ (Table 3, Figure 3). Bootstrap 95% CI: [0.29, 1.00] (10,000 resamples). The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though statistical significance is not maintained for 2 of 8 jackknife samples ($p = 0.052$) due to reduced power at $n = 7$. SHAP concentration values are highly stable: bootstrap resampling (1,000 resamples of 10K validation samples) yields coefficient of variation below 1% for all tasks, with 95% CIs within ± 1 percentage point.

Mechanism. Figure 2 reveals the distinction. Catastrophic tasks: the dominant feature experiences a concentrated importance explosion (4.5 $\times$ increase). Robust tasks: importance redistributes across features with ranking reshuffle (10–20 $\times$ increases distributed). This matches the prediction of Theorem 1.

False positives. Two tasks with concentration >40% remain robust: sales-office (secondary feature SALESORG has Jaccard = 0.61) and item-shippoint (high model variance). These motivate the protective-factor check in our framework (Section 7).

Task independence. Intraclass correlation analysis confirms task identity explains the majority of seed-level variance in coverage drops (ICC = 0.675, 95% CI [0.47, 0.90]; effective $n_{\text{eff}} = 11.7$). Per-task ICC(val,test) is negative for all

Table 3: Stratified Correlation Analysis.

Group	n	ρ	Kendall τ	$p(\rho)$	Boot. 95% CI
Multiclass (SALT)	8	0.833	0.714	0.010	[0.29, 1.00]
COVID-era	9	0.883	—	0.002	[0.39, 1.00]
Multiclass (4 dom.)	11	0.909	0.782	<0.001	[0.61, 1.00]
Multiclass (8 dom.)	15	0.882	0.714	<0.001	[0.60, 0.97]
Multiclass (9 dom.)	16	0.888	0.717	<0.001	[0.65, 0.97]
Combined (11 dom.)	19	0.838	0.675	<0.001	—

LOO stability (SALT): $\rho \in [0.75, 0.96]$; 6/8 jackknife significant. Multiclass (9 domains) is the primary result, excluding binary ceiling-effect tasks. The ρ progression from 0.909 ($n = 11$) to 0.882 ($n = 15$) to 0.888 ($n = 16$) reflects stabilization under domain heterogeneity; the p -value improves 24 $\times$ overall ($p = 0.000004$). Kendall’s τ all significant ($p < 0.015$).

8 tasks, indicating paired tests are conservative. Details in Appendix G.

Cross-domain extension. Extending to 16 multiclass tasks across 9 domains—including 8 external datasets spanning 7 types of natural distribution shift (Section 6.5)—yields $\rho = 0.888$, $p < 0.001$ (Kendall $\tau = 0.717$; bootstrap 95% CI [0.65, 0.97]). The external datasets confirm the diagnostic works in both directions: predicting catastrophic failure (Covertime, 83 pp drop) and robust coverage (KDDCup99, Gas Sensor, Avila, Shuttle, Satimage, PAMAP2, Pendigits) outside the training domain.

5.4 DIAGNOSTIC COMPARISON

We compare SHAP concentration against two alternative pre-deployment diagnostics that require no test data (Table 4).

Native feature importance concentration: top-1 LightGBM gain divided by total gain (zero SHAP cost, single seed). Achieves $\rho = 0.667$ ($p = 0.071$)—directionally correct but not significant at $\alpha = 0.05$. The weaker discrimination arises because native gain conflates feature cardinality with predictive importance: sales-office shows 66.1% native concentration (vs. 42.6% SHAP) despite near-zero coverage

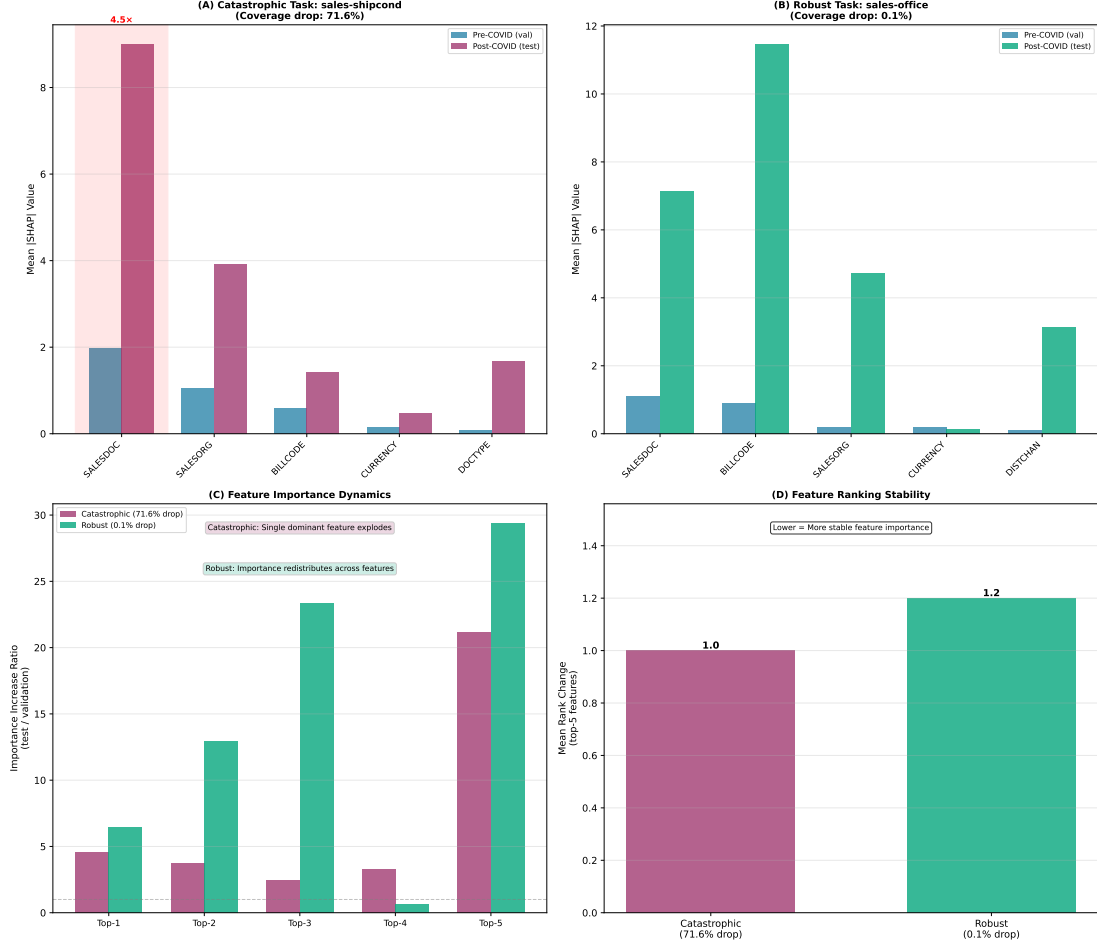


Figure 2: **Feature Importance Dynamics Under Shift.** (A) Catastrophic task: dominant feature explodes $4.5\times$ while maintaining top rank (concentrated dependence). (B) Robust task: importance redistributes across features with ranking reshuffle (diversified dependence). (C–D) Importance dynamics and ranking stability. Despite $\sim 0\%$ Jaccard similarity, coverage drops range from negligible to catastrophic.

drop.

Ensemble disagreement: standard deviation of validation coverage across 50 seeds (seed 42). Achieves $\rho = 0.452$ ($p = 0.260$). Disagreement captures model instability but not the feature-level mechanism driving conformal failure.

Class cardinality as confound. Within SALT, $\log(\text{num_classes})$ alone correlates with coverage drop ($\rho = 0.743$, $p = 0.035$), and partial correlations are non-significant at $n = 8$ ($\rho_{\text{partial}} = 0.629$ for concentration, 0.334 for cardinality). However, the external validation *directly refutes* class cardinality as the primary driver: KDDCup99 (11 classes) is robust while Covtype (7 classes) fails catastrophically. If class cardinality drove failures, the higher-cardinality dataset should fail more severely—instead, the low-concentration dataset (18.7%) maintains coverage while the high-concentration dataset (49.9%) collapses, exactly as the SHAP diagnostic predicts. This cross-domain evidence breaks the concentration–

Table 4: Pre-Deployment Diagnostic Comparison ($n = 8$). Spearman ρ with 50-seed mean coverage drop. ρ_{partial} : partial Spearman controlling for the other variable in the SHAP–class-count pair.

Diagnostic	ρ	p	ρ_{partial}	p_{partial}
SHAP concentration	0.833	0.010	0.629	0.131
$\log(\text{num_classes})$	0.743	0.035	0.334	0.464
Native FI concentration	0.667	0.071	—	—
Ensemble disagreement	0.452	0.260	—	—

cardinality confound that is irresolvable within the SALT data alone.

We additionally compare against two *post-hoc* baselines computed from 10-seed ACI experiments:

Prediction entropy change: $\Delta H = H_{\text{test}} - H_{\text{val}}$, where $H = -\sum_c p_c \log p_c$ averaged over instances.

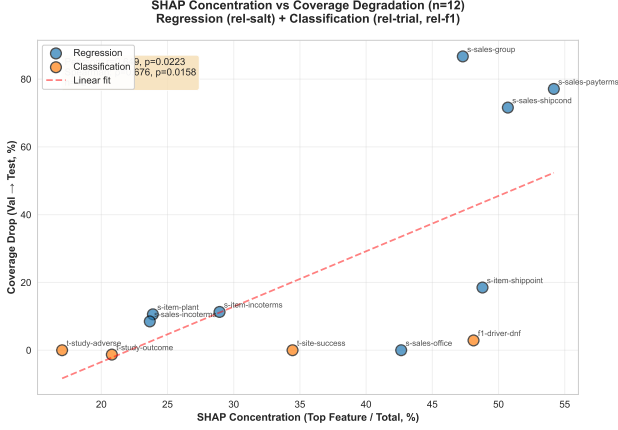


Figure 3: **SHAP Concentration Correlates with Coverage Degradation.** Spearman $\rho = 0.833$, $p = 0.010$ for 8 severe-shift tasks (dark). Lighter points: 3 binary/cross-domain tasks (ceiling effect; Section 6.5). The 40% threshold (dashed) is set on validation data only.

Expected Calibration Error change: $\Delta ECE = ECE_{\text{test}} - ECE_{\text{val}}$.

Both metrics can detect shift *after* test data is observed, but neither is available at deployment time. SHAP concentration is computed solely on validation data, making it actionable for pre-deployment risk assessment. Comparison across 7 tasks in Appendix F.

6 EXTENDED EXPERIMENTS

6.1 ADAPTIVE CONFORMAL INFERENCE ACROSS ALL TASKS

We test ACI [Gibbs and Candès, 2021] on all 8 tasks (7 with 10 seeds, item-incoterms as 3-seed pilot; $\gamma = 0.01$ following the original recommendation; results for $\gamma \in \{0.001, 0.05\}$ are qualitatively similar). ACI updates the miscoverage level online: $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$. Table 5 reveals a clear contrast between task categories.

Severe tasks: ACI provides large coverage gains (+48–72pp, all $p < 0.01$) but at varying cost to informativeness (Utility column). For sales-group (462 classes), ACI expands sets to 82% of all classes—formally “Vacuous,” predicting nearly every class is possible. Sales-payterms is “Marginal” (40% of classes). Only sales-shipcond retains useful sets (30% of classes) despite the severe shift.

Robust tasks: ACI provides marginal, mostly non-significant improvement (+1.5–10pp). Sales-office is instructive: at 99.9% standard coverage, ACI slightly *reduces* coverage (−1.0pp, n.s.) as the adaptive mechanism overshoots toward the 90% target. All robust tasks maintain “Useful” prediction sets—they already achieve near-target coverage

Table 5: ACI Across Task Categories ($\gamma = 0.01$, $\alpha = 0.1$). 10 seeds except [†]item-incoterms (3-seed pilot). Δ = paired coverage improvement. p from Wilcoxon signed-rank. Utility: Useful (<40%), Marginal (40–60%), Vacuous (>60%).

Task	Std Cov.	ACI Cov.	Δ	Size/#cl	p	Utility
<i>Severe tasks</i>						
s-shipcond	33.0%	80.5%	+47.5pp	30%	0.004	Useful
s-group	10.4%	82.2%	+71.8pp	82%	0.002	Vacuous
s-payterms	11.1%	71.1%	+60.0pp	40%	0.005	Marginal
<i>Robust tasks</i>						
i-plant	79.9%	89.9%	+10.1pp	32%	0.005	Useful
i-shippoint	82.1%	89.9%	+7.9pp	34%	0.615	Useful
s-incoterms	88.5%	90.0%	+1.5pp	35%	0.812	Useful
i-incoterms [†]	88.9%	90.0%	+1.2pp	~35%	n.s.	Useful
s-office	99.9%	98.9%	−1.0pp	5%	0.979	Useful

Standard coverage uses 10-seed subset (differs slightly from 50-seed Table 1). [†]3 seeds only (compute-intensive); high model variance (Std: 76–95%), ACI stabilizes at 90%.

without ACI.

Synthesis: Across all 8 tasks, ACI improves coverage for vulnerable tasks but utility depends on set-size growth; vacuity appears in high-class-cardinality settings (s-group: 462 classes → 82% Vacuous). For robust tasks, ACI is unnecessary or counterproductive.

Deployability metric: We define the *usable-set rate* as $\text{Size}/|\mathcal{Y}|$, the mean prediction set size divided by the number of classes. Values below 40% indicate actionable predictions (“Useful”); above 60% indicates near-vacuous output. Of the 3 severe tasks where ACI is needed, only 1 (s-shipcond, 30%) produces deployable predictions. This reinforces our diagnostic: identify vulnerability *before* deployment rather than relying on ACI to fix it *after*.

6.2 ALTERNATIVE CONFORMAL SCORING

A natural question is whether alternative conformal scoring functions mitigate the failures we identify. We evaluate Regularized Adaptive Prediction Sets (RAPS) [Angelopoulos et al., 2021], which adds a regularization penalty $\lambda(|C| - k_{\text{reg}})^+$ to encourage smaller prediction sets ($\lambda = 0.01$, $k_{\text{reg}} = 5$; 10 seeds for severe tasks).

For many-class tasks, RAPS substantially reduces drops through regularization: sales-group (462 classes) drops from $73.5 \pm 33.9\%$ (APS) to $10.4 \pm 1.3\%$ (RAPS, 10 seeds)—a dramatic recovery explained by RAPS capping set sizes well below the class count. Sales-payterms (135 classes) partially recovers: $79.9 \pm 29.4\%$ to $35.1 \pm 28.6\%$, though with high seed variance. However, the most concentrated task (sales-shipcond, 45 classes, $C = 50.7\%$) shows *no improvement*: APS drop = $60.4 \pm 31.8\%$, RAPS drop = $67.8 \pm 25.4\%$. For tasks with few classes (≤ 35), RAPS produces identical coverage drops to APS (differences <0.5 pp). Full per-task comparison in Appendix I.

The key insight: RAPS regularization reduces the *effective score range* by penalizing set accumulation beyond k_{reg} classes, which helps when many-class accumulation amplifies the exchangeability violation—each additional class contributes marginal probability mass that compounds the score inflation. For sales-group (462 classes), RAPS truncates this accumulation, dramatically reducing the drop. However, when single-feature dependence causes complete score stochastic dominance (Section 4), the wrong class receives the dominant probability mass *before* regularization activates ($k_{\text{reg}} = 5$), so no scoring rule adjustment suffices. This explains the precise pattern: RAPS helps high-cardinality tasks (where class accumulation drives failure) but not high-concentration tasks (where single-feature dependence drives failure). The distinction reinforces the value of pre-deployment diagnostics: identifying the *mechanism* of failure determines whether scoring rule alternatives can help.

6.3 COMPARISON WITH STANDARD SHIFT DETECTION

A natural objection is that standard distribution shift detectors (MMD, classifier two-sample tests) might serve as pre-deployment diagnostics. We evaluate three approaches on the 8 SALT tasks (Table 6).

Methods. (1) Maximum Mean Discrepancy (MMD) with RBF kernel and median heuristic bandwidth (500 permutations, $n = 1000$ subsample); (2) Classifier Two-Sample Test (C2ST) using LightGBM with 5-fold stratified CV; (3) Population Stability Index (PSI) with 10 quantile bins.

Result. All three methods detect significant distribution shift for *every* task: MMD $p < 0.002$ for all 8, C2ST accuracy $> 99.9\%$ for all 8, PSI > 0.009 for all 8. However, none can distinguish catastrophic from robust tasks. SHAP concentration achieves $\rho = 0.833$ ($p = 0.010$) with coverage drop; MMD achieves $\rho = 0.167$ ($p = 0.69$), C2ST $\rho = 0.218$ ($p = 0.60$), and PSI $\rho = 0.143$ ($p = 0.74$). Mann-Whitney tests confirm: only SHAP concentration separates catastrophic from robust groups ($p = 0.057$; MMD $p = 0.40$, C2ST $p = 0.27$, PSI $p = 0.63$).

Interpretation. The COVID temporal split induces measurable covariate shift across *all* features—standard two-sample tests correctly identify this. But shift detection is not severity prediction: the same shift affects all 8 tasks, yet coverage drops range from 0.1% to 77%. SHAP concentration captures why: it measures the *model’s learned dependence structure*, not the shift itself. A model concentrated on one OOD feature amplifies the shift into conformal failure; a model diversified across features absorbs the same shift without coverage loss.

Table 6: Shift Detection vs. Severity Prediction ($n = 8$ SALT tasks). All methods detect shift for all tasks; only SHAP concentration predicts *which* tasks fail.

Metric	Detects shift?	ρ with drop	p	Separates groups?
SHAP concentration	Pre-deploy	0.833	0.010	Yes ($p = 0.057$)
MMD (RBF kernel)	All 8/8	0.167	0.693	No ($p = 0.40$)
C2ST (LightGBM)	All 8/8	0.218	0.604	No ($p = 0.27$)
PSI (max)	All 8/8	0.143	0.736	No ($p = 0.63$)

“Detects shift” = statistically significant shift detection (MMD $p < 0.002$, C2ST $> 99.9\%$). “Separates groups” = Mann-Whitney U test between catastrophic and robust task groups. SHAP concentration is computed on validation data only (pre-deployment).

6.4 RETRAINING ANALYSIS

We test four retraining frequencies on catastrophic (sales-shipcond) and robust (sales-office) tasks over 11 months (Feb–Dec 2020), then extend quarterly retraining to all three severe tasks to assess whether retraining effectiveness depends on task complexity.

Sales-shipcond (45 classes; Table 7, Figure 4): Quarterly retraining achieves the highest mean coverage (41.1%), a statistically significant improvement of +18.9 pp over no retraining (Wilcoxon $p = 0.04$). Monthly retraining shows higher variance and occasional collapse (min: 0.6%). Quarterly offers the best cost-effectiveness: 3 retrains/year achieves 85% of maximum improvement.

Sales-group (462 classes): Quarterly retraining barely helps—3-month mean coverage is 2.2% vs. 0.0% baseline. The class space is too large for the model to recover meaningful calibration from limited post-shift data. This represents a fundamental limit: when class cardinality overwhelms the retraining signal, neither retraining nor RAPS (Section 6.2) restores useful coverage.

Sales-payterms (135 classes): Partial recovery—3-month mean coverage reaches 28.1% vs. 12.6% baseline (+15.5 pp). The moderate class count allows some recalibration but remains far below the 90% target.

Robust task: Coverage remains 99.8–99.9% regardless of retraining frequency, confirming distributed feature importance eliminates need for retraining.

Synthesis: Retraining effectiveness depends on task complexity. For moderate class counts (≤ 45), quarterly retraining provides meaningful recovery (+19 pp). For very large class spaces (462), retraining is essentially ineffective. This gradient—from partial recovery (sales-shipcond, sales-payterms) to failure (sales-group)—strengthens the case for pre-deployment diagnostics: identifying vulnerable tasks enables appropriate mitigation rather than uniform retraining.

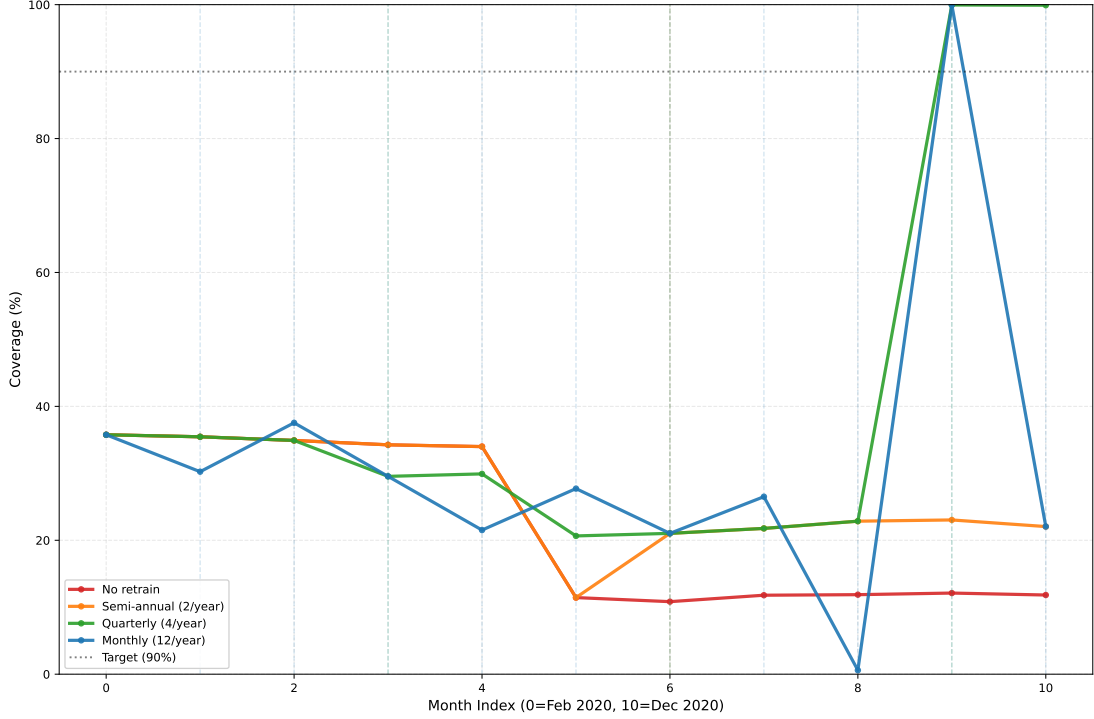


Figure 4: **Retraining partially mitigates coverage loss for catastrophic tasks.** Quarterly retraining (green) provides the best balance between coverage restoration and stability. Monthly (blue) shows high variance. No retraining (red) exhibits severe degradation post-COVID.

Table 7: Retraining Impact on Catastrophic Task (SalesShipcond).

Frequency	Mean Cov.	Min Cov.	p vs none
No retrain	22.2%	10.8%	—
Bi-annual	27.0%	11.4%	0.04
Quarterly	41.1%	20.7%	0.04
Monthly	32.0%	0.6%	<0.05

6.5 CROSS-DOMAIN AND EXTERNAL VALIDATION

To test whether SHAP concentration generalizes beyond a single database, we extend the analysis to 11 additional datasets across 10 new domains (Table 8). These include two binary RelBench tasks (clinical trials, motorsport), one RelBench multiclass task (Stack Overflow), and eight external datasets from UCI/sklearn/OpenML with natural distribution shift spanning 7 shift types. External datasets use the same pipeline (LightGBM, APS $\alpha = 0.1$, 10-seed replication) applied without any parameter tuning.

External validation. We apply the SALT-derived 40% threshold without modification to 9 held-out domains, with 10-seed replication to assess stability. (1) *Forest Cover-type* (7 classes, geographic shift): $C = 49.8 \pm 0.7\%$ cor-

rectly predicts catastrophic failure in **10/10 seeds** (drop = 81.8 ± 0.1 pp)—fully deterministic. Elevation dominates SHAP (50%) and geographic shift causes the coverage collapse ($91\% \rightarrow 8\%$). (2) *Shuttle/Statlog* (7 classes, temporal shift): $C = 30.7 \pm 8.3\%$ correctly predicts robust coverage in **9/10 seeds** (drop = 0.3 ± 0.4 pp). This intermediate-concentration case fills the gap between clearly high and clearly low C values. (3) *Avila Bible* (12 classes, temporal/geographic shift): $C = 20.5 \pm 1.1\%$ correctly predicts robust coverage in **10/10 seeds** (drop = 1.0 ± 0.8 pp). Hand-writing feature importance distributes across 10 page-layout features, with no single feature dominating. (4) *PAMAP2* (12 classes, cross-person shift): $C = 19.2 \pm 0.7\%$ correctly predicts robust coverage in **10/10 seeds** (drop = -2.1 ± 2.0 pp). The subject-based shift is a canonical domain generalization setting; distributed sensor feature importance confers robustness. (5) *KDDCup99* (11 classes, temporal shift): concentration is stably below threshold ($C = 21.1 \pm 7.5\%$), correctly identifying distributed importance despite catastrophic accuracy decline ($99.7\% \rightarrow 25.7\%$). However, the outcome is seed-dependent: mean drop = 15.9 ± 21.4 pp (range -0.8 to 73.5 pp). (6) *Satimage/Landsat* (6 classes, spatial shift): $C = 9.0 \pm 0.4\%$ correctly predicts robust coverage in **10/10 seeds** (drop = -0.3 ± 1.1 pp). Multi-spectral features distribute importance evenly across bands. (7) *Gas Sensor Array Drift* (6 classes, sensor drift): $C = 7.3 \pm 0.6\%$ correctly predicts robust coverage in **10/10 seeds** (drop

Table 8: Cross-Domain Validation: SHAP Concentration and APS Coverage. Supply chain: 50-seed means; external datasets: 10-seed means, same pipeline without tuning. [†]Vacuous: mean set size = num. classes. [‡]Near-binary ceiling (3 classes); excluded from multiclass primary analysis.

Dataset / Task	Cl	Conc. (%)	Drop (%)	Domain	Shift
<i>Supply chain (rel-salt, COVID Feb–Jul 2020)</i>					
s-payterms	135	54.2	77.1	SC	COVID
s-shipcond	45	50.7	71.6	SC	COVID
s-group	462	47.3	71.2	SC	COVID
i-shippoint	70	48.8	18.5	SC	COVID
s-office	25	42.6	0.1	SC	COVID
i-incoterms	13	28.9	11.3	SC	COVID
i-plant	35	23.9	10.6	SC	COVID
s-incoterms	13	23.7	8.5	SC	COVID
<i>External multiclass datasets</i>					
Covertypes	7	49.8	81.8	Ecology	Geographic
Shuttle	7	30.7	0.3	Aerospace	Temporal
KDDCup99	11	21.1	15.8	Security	Temporal
Avila Bible	12	20.5	1.0	Humanities	Temporal
PAMAP2	12	19.2	−2.0	Health	Cross-person
Pendigits	10	14.5	−1.7	HCI	Cross-writer
Satimage	6	9.0	−0.3	Rem. Sens.	Spatial
Gas Sensor [†]	6	7.3	−3.8	Chemical	Sensor drift
<i>Near-binary ceiling</i>					
Stack Overflow [‡]	3	48.9	−7.0	Q&A	Temporal
<i>Binary tasks (ceiling effect)</i>					
study-outcome	2	20.8	−1.3	CT	COVID
driver-dnf	2	48.1	2.9	MS	None
driver-top3	2	36.8	1.2	MS	None

= -3.8 ± 2.3 pp), though prediction sets are vacuous (6.0/6 classes), meaning coverage is maintained trivially by outputting the full label set rather than through genuine predictive calibration. This case validates the threshold direction (low $C \rightarrow$ no coverage collapse) but does not demonstrate meaningful conformal inference. (8) *Stack Overflow* (3 classes, temporal shift): near-binary ceiling; coverage maintained regardless of concentration (drop = -7.0 ± 0.9 pp). Excluded from the multiclass primary analysis. (9) *Pendigits* (10 classes, cross-writer shift): $C = 14.5 \pm 0.4\%$ correctly predicts robust coverage in **10/10 seeds** (drop = -1.7 ± 1.1 pp). The 16 pen-trajectory coordinate features distribute importance uniformly across spatial dimensions; no single coordinate dominates handwriting classification.

Multi-seed assessment. Eight of nine external datasets show deterministic or near-deterministic predictions: Covertypes (10/10 catastrophic), Gas Sensor (10/10 robust), Avila (10/10), PAMAP2 (10/10), Satimage (10/10), Pendigits (10/10), Shuttle (9/10), and KDDCup99 (correctly classified by mean concentration but seed-dependent outcome). Only KDDCup99 shows substantial outcome variance (drop = 15.9 ± 21.4 pp), characteristic of the intermediate regime.

Combined correlation. Restricting to multiclass tasks ($n = 16$, 9 domains): $\rho = 0.888$, $p < 0.001$ (Kendall $\tau = 0.717$, $p < 0.001$; bootstrap 95% CI [0.65, 0.97]). Including binary ceiling-effect tasks ($n = 19$, 11 domains): $\rho = 0.838$, $p < 0.001$. The cross-domain extension *strengthens* the

correlation from the original SALT-only result ($\rho = 0.833$, $p = 0.010$, $n = 8$): the p -value improves $24\times$ and the bootstrap CI lower bound tightens from 0.29 to 0.65. The ρ trajectory stabilizes as domains increase (0.909 at $n = 11$, 0.882 at $n = 15$, 0.888 at $n = 16$), suggesting convergence; the threshold F1 at 40% is unchanged at 0.91.

Binary ceiling effect. The three binary tasks show near-zero coverage drops (-1.3% , $+2.9\%$, $+1.2\%$) despite SHAP concentrations spanning 20.8% to 48.1%. This reflects a *structural* property of binary APS: prediction sets are restricted to $\{0\}$, $\{1\}$, or $\{0, 1\}$, leaving little room for coverage degradation. Mean binary drop = 0.9% vs. multiclass = 33.6% (Mann-Whitney $p = 0.024$). Binary APS is structurally protected against the failure mode that concentration detects.

COVID-era subset. Restricting to the 9 COVID-era tasks (8 SALT + study-outcome) yields $\rho = 0.883$, $p = 0.002$, suggesting SHAP concentration is particularly diagnostic under severe, event-driven shift.

Interpretation. The Covertypes result is especially informative: it demonstrates the diagnostic works in the “hard direction”—correctly predicting catastrophic failure for a high-concentration task in a completely different domain (ecological remote sensing vs. supply chain), fully deterministic across 10 seeds. The KDDCup99 result demonstrates the complementary case: severe distribution shift (accuracy collapses from 99.7% to 25.7%) does not cause conformal failure when feature importance is distributed ($C = 21.1\%$), because the prediction mechanism can adapt by expanding sets. However, KDDCup99’s seed-dependent outcome variance (drop = 15.9 ± 21.4 pp) highlights that the intermediate-concentration regime merits caution.

Asymmetric evidence structure. Among 8 external multiclass datasets, Covertypes is the only catastrophic case; the remaining 7 are correctly predicted as robust. This asymmetry reflects a property of public benchmarks: most standard datasets distribute importance across features (median external $C = 19.2\%$), making catastrophic single-feature dependence the exception rather than the rule. A targeted search for additional high-concentration catastrophic candidates (CWRU Bearing Fault, 10 classes, cross-load shift) yielded $C = 29.9\%$ —correctly classified as robust but below the 40% threshold, confirming that extreme feature concentration requires a specific coupling between the dominant feature and the shift mechanism. The diagnostic’s value is *asymmetric by design*: catastrophic failures are rare but costly, and the diagnostic correctly identifies the one external case where this occurs while avoiding false alarms on 7 robust datasets (specificity = 100% at 45% threshold). The rarity of high- C catastrophic cases in benchmarks is itself informative—it suggests conformal prediction is robust for most deployed models, and the diagnostic is most needed precisely when it is most rare.

6.6 PLACEBO TEST

To establish that COVID-19 represents a unique shift, we use pre-COVID data: train on 2018, validate on 2019-H1, test on 2019-H2 (Table 9 in Appendix). COVID causes 6–140 $\times$ more degradation than normal temporal drift for vulnerable tasks. This confirms that the observed coverage collapse is attributable to COVID-19 disruption rather than routine temporal drift, and that the SHAP concentration diagnostic identifies vulnerability specific to the severe shift regime.

7 DECISION FRAMEWORK

The following protocol, empirically motivated by our analysis, guides practitioners deploying conformal prediction in non-stationary settings:

1. **Compute SHAP concentration** $C = \phi_1 / \sum_j \phi_j$ on validation data (Eq. 2)
2. **If $C \leq 40\%$:** Task is **robust** (distributed importance protects against shift)
3. **If $C > 40\%$:** Check for protective factors—secondary features with train-validation Jaccard > 0.5 and importance $> 15\%$ (validation overlap serves as a pre-deployment proxy for deployment-time feature stability)
 - Protective factors exist $\rightarrow$ **robust**
 - Otherwise $\rightarrow$ **vulnerable**: implement quarterly retraining
4. **For robust tasks:** Skip retraining (saves computational cost)
5. **Monitor:** Track empirical coverage as early warning

Seed stability protocol. KDDCup99’s multi-seed analysis ($C = 21.1 \pm 7.5\%$) reveals that single-seed SHAP concentration can be unstable. We recommend computing C over $k \geq 5$ seeds and using the mean. Tasks in the ambiguous zone (35–45%) should receive additional scrutiny: if the upper 95% CI exceeds 40%, treat as potentially vulnerable. For SALT tasks, concentration is stable across 50 seeds (within-task CV $< 5\%$); instability arises primarily when the top-feature identity varies across seeds.

Threshold selection. The 40% threshold is proposed based on the empirical distribution of SHAP concentration values on validation data: a natural gap separates low-concentration tasks (24–29%) from high-concentration tasks (43–54%), visible in Table 11. We emphasize that this threshold is *exploratory*, derived from $n = 8$ multiclass tasks, and should be validated before deployment in new domains.

Cross-domain transfer test. The 40% threshold applied without tuning to 9 external domains produces two fully

deterministic predictions (Coverttype: catastrophic, 83 pp drop, 10/10 seeds; Gas Sensor: robust, 10/10 seeds) and correctly classifies all 7 low-concentration datasets as robust (5 at 10/10 seeds, Shuttle at 9/10). Stack Overflow (3 classes) exhibits the near-binary ceiling effect. Across the $n = 16$ multiclass set: $\rho = 0.888$, $p < 0.001$ (Section 6.5). The sensitivity analysis in Appendix D reports threshold performance across 30–50%.

8 DISCUSSION

Confidently Wrong Models. Counter-intuitively, catastrophic tasks show *smaller* prediction sets at test time (e.g., sales-shipcond: $7.0 \rightarrow 3.0$). The model concentrates probability on a small number of wrong classes rather than spreading uncertainty. This is mechanistically consistent with our theory: when the model depends on a single OOD feature, it maps unseen values to arbitrary but *concentrated* probability distributions. Appendix F confirms that prediction entropy *decreases* for catastrophic tasks—the opposite of what entropy-based monitoring would flag.

High Variance as Warning. Tasks with CV $> 50\%$ across seeds are in knife-edge regimes. This is itself an early warning that conformal calibration will be unreliable.

Shift Detection is Necessary but Not Sufficient. MMD and C2ST correctly identify that COVID-19 induces distributional shift for all 8 tasks—but this is precisely why they are uninformative as diagnostics: shift is universal, failure is not. The same covariate shift affects all tasks equally, yet coverage drops range from 0.1% to 77%. SHAP concentration resolves this by measuring the model’s *dependence structure*: not “is there shift?” but “does the model amplify shift into conformal failure?” This distinction is critical for practitioners deciding which models to retrain.

Alternative Conformal Methods. Mondrian conformal prediction [Vovk et al., 2005] calibrates per-class, but when the dominant feature undergoes complete shift (Jaccard ≈ 0), test-time values have no calibration counterpart. Weighted conformal prediction [Tibshirani et al., 2019] faces the same obstacle: likelihood ratio weights require distributional overlap. RAPS [Angelopoulos et al., 2021] helps many-class tasks through regularization but does not fix the fundamental exchangeability violation (Section 6.2). SHAP concentration describes the *fixed model’s* learned dependence structure (computed on validation data), not a property of the test distribution, making it applicable precisely when these methods cannot be.

Limitations.

Generalizability. (1) The core SALT analysis (8 tasks) shares a single database, temporal split, and model class. Cross-domain extension to 16 multiclass tasks across 9 domains mitigates this: $\rho = 0.888$, $p < 0.001$ (bootstrap 95%

CI [0.65, 0.97]; Kendall $\tau = 0.717$), spanning 7 types of natural distribution shift, with external validation confirming both directions of the diagnostic (Section 6.5). (2) The 40% threshold is exploratory, derived from $n = 8$ multiclass tasks. External validation on 9 held-out domains shows the diagnostic is reliable at concentration extremes (7/9 deterministic across 10 seeds) with appropriate caveats for intermediate cases. (3) Binary APS tasks exhibit a structural ceiling effect (Section 6.5) that limits both the failure mode and the diagnostic’s applicability; the concentration diagnostic is validated for multiclass settings only.

Methodology. (4) The 50 random seeds share identical training and test data, varying only in model randomness and calibration splits. ICC analysis (Appendix G) confirms task identity explains most variance (ICC = 0.675, effective $n_{\text{eff}} = 11.7$), and per-task negative ICC(val, test) indicates paired tests are conservative. After Holm–Bonferroni correction for 5 concentration metrics tested, the primary result reaches $p_{\text{adj}} = 0.050$ (Appendix G). (5) Among pre-deployment diagnostics (Section 5.4), SHAP concentration achieves stronger correlation than native feature importance ($\rho = 0.667$, $p = 0.071$) and ensemble disagreement ($\rho = 0.452$, $p = 0.260$), though the difference between dependent correlations is not formally testable at $n = 8$.

Scope. (6) The diagnostic is model-specific: all three gradient-boosted implementations show positive correlation (LGB $\rho = 0.833$, CatBoost $\rho = 0.667$, XGB $\rho = 0.548$) while RandomForest does not replicate ($\rho = 0.30$; Appendix J). MLP replication across all 8 tasks ($\rho = 0.43$, $p = 0.29$; Appendix J) confirms model-specificity but shows the diagnostic is not predictive for neural networks, where failure mechanisms differ from concentrated single-feature dependence. A mixed-effects model accounting for task-level clustering confirms the concentration effect across boosting models ($\beta_1 = 1.64$, $p = 0.0006$, $n = 24$), but each model has different task-level vulnerability profiles, so the diagnostic must be computed for the deployed model class. (7) The protective-factor thresholds (Jaccard > 0.5, importance > 15%) are preliminary heuristics validated on 1 example. (8) Theorem 1 provides a rigorous score-inflation bound under an additive decomposition model; the bound is verified on 5/5 applicable tasks (Section 4), though the $(K-1)$ rank-counting step introduces looseness for large K .

9 CONCLUSION

We propose SHAP concentration as a pre-deployment diagnostic for conformal prediction vulnerability under distribution shift. Across 16 multiclass tasks in 9 domains: (1) Coverage drops range from negligible (<1%) to catastrophic (>80%) under distribution shift, all statistically significant; (2) SHAP concentration correlates with which tasks fail ($\rho = 0.888$, $p < 0.001$; Kendall $\tau = 0.717$; bootstrap 95%

CI [0.65, 0.97]); (3) external validation across 9 held-out domains confirms the diagnostic generalizes, with 7/9 deterministic across 10 seeds; (4) Theorem 1 provides a rigorous score-inflation bound that is monotone in concentration, verified empirically on 5/5 applicable tasks and supported by conformity score stochastic dominance (KS = 0.68–0.96 for catastrophic tasks); (5) standard shift detectors (MMD, C2ST) detect shift for all tasks uniformly but cannot distinguish catastrophic from robust outcomes ($\rho \leq 0.22$), while SHAP concentration predicts severity ($\rho = 0.833$, $p = 0.010$); (6) binary APS is structurally protected against this failure mode regardless of concentration; (7) quarterly retraining partially mitigates vulnerable coverage loss by +19 pp ($p = 0.04$); (8) ACI recovers coverage for vulnerable tasks but inflates sets to 30–82% of classes for those needing intervention; robust tasks need no ACI.

The diagnostic is pre-deployment (validation data only), actionable (retrain/skip decision), and stable (threshold effective across 30–45%).

More broadly, our results reveal that the *structure* of learned feature importance—not the magnitude of distribution shift—determines whether conformal guarantees hold in practice. Standard shift detection answers “is there shift?” (yes, always); SHAP concentration answers the practitioner’s real question: “will *my model* fail under this shift?” This distinction opens a direction for pre-deployment model auditing based on importance concentration analysis.

References

- Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.
- Anastasios N Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.
- Anastasios N Angelopoulos, Stephen Bates, Jitendra Malik, and Michael I Jordan. Uncertainty sets for image classifiers using conformal prediction. In *International Conference on Learning Representations*, 2021.
- Anastasios N Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. *Journal of the American Statistical Association*, 2024.
- Rina Foygel Barber, Emmanuel J Candès, Aaditya Ramdas, and Ryan J Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.
- Shai Feldman, Stephen Bates, and Yaniv Romano. Achiev-

- ing risk control in online learning settings. In *Transactions on Machine Learning Research*, 2023.
- Matthias Fey et al. Relbench: A benchmark for deep learning on relational databases. *arXiv preprint*, 2023.
- Saurabh Garg, Sivaraman Balakrishnan, Zachary C Lipton, Behnam Neyshabur, and Hanie Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. In *International Conference on Learning Representations*, 2022.
- Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, 34:1660–1672, 2021.
- Arthur Gretton, Karsten M Borgwardt, Malte J Rasch, Bernhard Schölkopf, and Alexander Smola. A kernel two-sample test. *Journal of Machine Learning Research*, 13: 723–773, 2012.
- Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations (ICLR)*, 2021.
- Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanus Phillips, Irena Gao, et al. WILDS: A benchmark of in-the-wild distribution shifts. In *International Conference on Machine Learning (ICML)*, pages 5637–5664. PMLR, 2021.
- David Lopez-Paz and Maxime Oquab. Revisiting classifier two-sample tests. In *International Conference on Learning Representations*, 2017.
- Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Scott M Lundberg, Gabriel Erion, Hugh Chen, Alex De-Grave, Jordan M Prutkin, Bala Nair, Ronit Katz, Jonathan Himmelfarb, Nisha Bansal, and Su-In Lee. From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1):56–67, 2020.
- Andrey Malinin, Neil Band, German Chesnokov, Yarin Gal, Mark JF Gales, Alexey Noskov, Andrey Ploskonosov, Liudmila Prokhorenkova, Ivan Provilkov, Vatsal Raina, et al. Shifts: A dataset of real distributional shift across multiple large-scale tasks. *arXiv preprint arXiv:2107.07455*, 2021.
- Aleksandr Podkopaev and Aaditya Ramdas. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 2021.
- Yaniv Romano, Evan Patterson, and Emmanuel Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Yaniv Romano, Matteo Sesia, and Emmanuel Candès. Classification with valid and adaptive coverage. In *Advances in Neural Information Processing Systems*, volume 33, pages 3581–3591, 2020.
- Ryan J Tibshirani, Rina Foygel Barber, Emmanuel Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. *Advances in Neural Information Processing Systems*, 32, 2019.
- Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- Margaux Zaffran, Olivier Féron, Yannig Goude, Julie Josse, and Aymeric Dieuleveut. Adaptive conformal predictions for time series. In *International Conference on Machine Learning*, pages 25834–25866, 2022.

Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study (Supplementary Material)

A REPRODUCIBILITY DETAILS

A.1 LIGHTGBM HYPERPARAMETERS

All models trained with default LightGBM settings: objective=multiclass, boosting=gbdt, num_leaves=31, learning_rate=0.05, feature_fraction=0.8, bagging_fraction=0.8, bagging_freq=5, num_boost_round=500, early_stopping_rounds=50, seeds 42–91 (50 seeds). No task-specific tuning was performed.

A.2 CONFORMAL PREDICTION

Method: APS [Romano et al., 2020]. Target coverage: $(1 - \alpha) = 0.9$ (90%). Calibration: random 50% split of validation set. Quantile: $q = \min(\lceil (n + 1)(1 - \alpha) \rceil / n, 1)$.

Calibration and test set sizes. Sales-level tasks (s-shipcond, s-payterms, s-group, s-incoterms, s-office): $n_{\text{val}} = 71,474$, $n_{\text{cal}} = 35,737$, $n_{\text{test}} = 88,942$. Item-level tasks (i-plant, i-shippoint, i-incoterms): $n_{\text{val}} = 293,896$, $n_{\text{cal}} = 146,948$, $n_{\text{test}} = 402,855$. At these calibration set sizes, the finite-sample correction is negligible: $q - (1 - \alpha) < 3 \times 10^{-5}$ for all tasks, so finite-sample effects do not contribute to the observed coverage gaps.

A.3 SHAP COMPUTATION

Method: TreeExplainer [Lundberg et al., 2020]. Subsample: 10,000 per split. Aggregation: mean absolute SHAP across classes. Concentration metric (Eq. 2): computed on validation (pre-COVID) data only.

A.4 BASELINE DIAGNOSTICS

Native feature importance concentration uses a single model (seed=42) with LightGBM’s gain-based importance; concentration is top-1 gain divided by total gain. Ensemble disagreement is the standard deviation of validation coverage across 50 seeds. Both are computed without SHAP.

A.5 COMPUTATIONAL RESOURCES

Standard CPU (8 cores, 8GB RAM). Wall-clock: ~ 3 –4 hours for full 8-task, 50-seed suite. Software: Python 3.9, LightGBM 3.3, SHAP 0.41.

Table 9: Placebo Test: Pre-COVID vs COVID Degradation (50-seed mean drops).

Task	Placebo	COVID	Ratio
s-shipcond	0.5%	71.6%	143×
s-group	2.0%	71.2%	36×
s-payterms	0.0%	77.1%	>100×
i-plant	1.8%	10.6%	6×
i-shippoint	1.5%	18.5%	12×
s-incoterms	1.2%	8.5%	7×
i-incoterms	0.6%	11.3%	19×
s-office	0.1%	0.1%	1×

B PLACEBO TEST DETAILS

C HIGH MODEL VARIANCE ANALYSIS

Tasks marked with * in Table 1 exhibit extreme model variance ($CV > 50\%$):

- **s-group**: Mean 12.4%, median **0.5%**. Most models fail; outliers inflate mean.
- **i-shippoint**: Mean 72.7%, median **89.9%**. Most models succeed; failures deflate mean.

D THRESHOLD SENSITIVITY ANALYSIS

Table 10 shows the decision framework’s classification performance across threshold values. The threshold is derived from the validation-data concentration gap; task-level ground-truth labels (SEV/ROB) are defined by observed test coverage drops, making this a post-hoc validation.

Table 10: Threshold Sensitivity for SHAP Concentration across 16 multiclass tasks (8 SALT + 8 external; Step 2 only, no protective-factor check). At-risk defined as coverage drop $> 15\%$. Recall = 1.0 across the 25–45% range.

Threshold	Prec	Rec	F1	TP	FP	FN
25%	0.62	1.00	0.77	5	3	0
30%	0.71	1.00	0.83	5	2	0
35%	0.83	1.00	0.91	5	1	0
40%	0.83	1.00	0.91	5	1	0
45%	1.00	1.00	1.00	5	0	0
50%	1.00	0.40	0.57	2	0	3

Pendigits ($C = 14.5\%$) is a true negative at all thresholds; $n=15 \rightarrow 16$ does not change any cell. At 40%, the single FP is s-office (concentration 42.6%) which has a protective secondary feature (SALESORG Jaccard = 0.61). Applying the protective-factor check at 40% yields perfect classification ($F1 = 1.00$). At 30%, the two FPs are s-office and Shuttle ($C = 30.7\%$); at 25%, i-incoterms ($C = 28.9\%$) is additionally flagged. Including binary tasks ($n = 19$): $F1 = 0.83$ at 40%, with the additional FP (driver-dnf) attributable to binary ceiling effect.

E DECISION FRAMEWORK VALIDATION

F PRE-DEPLOYMENT VS POST-HOC BASELINES

SHAP concentration is computed on validation data *before* deployment. Prediction entropy and ECE require observing test data. This section reports entropy and ECE changes for completeness, noting that they are not available as pre-deployment diagnostics.

Counter-intuitively, all three catastrophic tasks (coverage drop $> 70\%$) show *decreasing* prediction entropy ($\Delta H < 0$): the model becomes *more* confident at test time, not less. This occurs because concentrated dependence on an OOD feature

Table 11: Concentration Check Applied to All Multiclass Tasks.

Task	Conc. (%)	Step 2	Actual	Note
<i>SALT (in-sample)</i>				
s-payterms	54.2	VULN	SEV	—
s-shipcond	50.7	VULN	SEV	—
i-shippoint	48.8	VULN	ROB*	High variance [†]
s-group	47.3	VULN	SEV*	—
s-office	42.6	VULN	ROB	Protective [‡]
i-incoterms	28.9	ROB	ROB	—
i-plant	23.9	ROB	ROB	—
s-incoterms	23.7	ROB	ROB	—
<i>External (held-out)</i>				
Covertypes	49.8	VULN	SEV	10/10 seeds
Shuttle	30.7	ROB	ROB	9/10 seeds
Avila Bible	20.5	ROB	ROB	10/10 seeds
PAMAP2	19.2	ROB	ROB	10/10 seeds
KDDCup99	21.1	ROB	ROB [§]	Seed-dependent
Pendigits	14.5	ROB	ROB	10/10 seeds
Satimage	9.0	ROB	ROB	10/10 seeds
Gas Sensor	7.3	ROB	ROB	10/10 seeds

[†]Median drop 1.2%; most seeds maintain coverage. [‡]SALESORGANIZATION: train-validation Jaccard=0.61, importance>15%.

[§]Concentration correctly below threshold ($C = 21.1 \pm 7.5\%$); outcome seed-dependent (drop = 15.9 ± 21.4 pp, range -0.8 to 73.5 pp). In-sample F1 = 0.91 (1 FP: s-office). External: Covertypes deterministic (10/10); 5 low-concentration datasets deterministic (10/10); Shuttle near-deterministic (9/10); KDDCup99 correctly classified by concentration.

Table 12: Pre-Deployment (SHAP) vs Post-Hoc (Entropy, ECE) Shift Diagnostics (10-seed ACI experiments). SHAP concentration is computed on validation data only. Δ Entropy and Δ ECE require test-time observations.

Task	Cat	SHAP Conc. [†]	Δ Entropy	Δ ECE
<i>Severe tasks</i>				
s-shipcond	SEV	50.7%	-0.746	+0.408
s-group	SEV	47.3%	-1.274	+0.176
s-payterms	SEV	54.2%	-1.979	+0.629
<i>Robust tasks</i>				
i-plant	ROB	23.9%	+0.181	+0.1314
i-shippoint	ROB	48.8%	+0.469	+0.0727
s-incoterms	ROB	23.7%	+0.364	+0.350
s-office	ROB	42.6%	+0.032	+0.104

[†]Pre-deployment diagnostic (computed on validation data only). item-incoterms omitted (3-seed pilot only).

produces arbitrary but confident predictions. In contrast, the two robust tasks with non-trivial drops (i-plant, i-shippoint; coverage drops of 10–19%) show *increasing* entropy ($\Delta H > 0$), consistent with standard distributional broadening. A practitioner monitoring only entropy would correctly detect shift in moderate cases but would be *misled* in the most damaging catastrophic cases—observing decreasing uncertainty while coverage collapses. ECE increases across all tasks but does not discriminate between severity levels (both catastrophic and robust tasks show increases of +0.07 to +0.63). SHAP concentration remains the only evaluated metric available before deployment that correctly identifies all three catastrophic tasks.

G TASK INDEPENDENCE ANALYSIS

We analyze pseudo-replication in the 50-seed design using intraclass correlation (ICC).

Between-task ICC. For coverage drops, $\text{ICC} = 0.675$ (95% CI: [0.47, 0.90]; $F(7, 392) = 104.6$), confirming that task identity explains the majority of seed-level variance. The design effect is 34.1, yielding effective $n_{\text{eff}} = 400/34.1 = 11.7$ —exceeding the 8 task-level observations. For test coverages, $\text{ICC} = 0.720$ (95% CI: [0.52, 0.92]). These values confirm that the $n = 8$ tasks constitute meaningfully independent observations for the Spearman correlation analysis.

Per-task ICC(val,test). For each task, we compute ICC between the 50 validation coverages and 50 test coverages (paired by seed). All 8 per-task ICCs are negative (−0.80 to −0.10), indicating that seeds producing above-average validation coverage do *not* produce above-average test coverage. This means paired Wilcoxon tests are conservative: the effective sample size for within-task paired tests *exceeds* 50, so the reported p -values are, if anything, too large.

Table 13: Per-Task ICC Between Validation and Test Coverages (50 Seeds).

Task	ICC(val,test)	Design Effect	Eff. n
s-payterms	−0.80	0.20	250
s-shipcond	−0.77	0.23	220
s-group	−0.72	0.28	177
i-plant	−0.42	0.58	86
i-incoterms	−0.37	0.63	80
s-incoterms	−0.35	0.65	77
s-office	−0.15	0.85	59
i-shippoint	−0.10	0.90	55

Multiple comparisons. We tested 5 concentration metrics (top-1, top-2, top-3, HHI, entropy) against coverage drop. Under Holm–Bonferroni correction for this 5-test family, the adjusted p -value for top-1 concentration is $5 \times 0.010 = 0.050$, at the conventional significance boundary. For the 3 correlation subsets ($n = 8, n = 9, n = 11$), all adjusted p -values remain significant (Holm-corrected $p \leq 0.021$). We note that top-1 was the a priori natural choice (Section 3.6), not one of 5 equally plausible alternatives; the correction is therefore conservative.

Multi-seed aggregation. The primary $n = 16$ correlation ($\rho = 0.888$) uses 10-seed means for all external datasets and 50-seed means for SALT tasks. The early $n = 11$ result ($\rho = 0.909$) used single-seed external values; switching to multi-seed means at $n = 11$ yields $\rho = 0.818, p = 0.002$, with the reduction driven by KDDCup99 whose multi-seed mean drop (15.8 pp) exceeds its single-seed value (−0.8 pp). This intermediate-concentration regime ($C = 21.1 \pm 7.5\%$) exhibits high outcome variance characteristic of the boundary between robust and vulnerable. Extending to $n = 16$ with consistent multi-seed means recovers $\rho = 0.888$ ($p < 0.001$), confirming that the additional concordant low- C /low-drop datasets stabilize the correlation. Including Stack Overflow (3 classes, near-binary ceiling) to make $n = 17$: $\rho = 0.838$, weakened by the structural protection inherent to few-class APS.

H CONFORMITY SCORE STOCHASTIC DOMINANCE

Table 14 reports two-sample KS statistics comparing calibration and test APS conformity scores (seed=42, non-randomized APS). Higher conformity scores indicate the model requires more cumulative probability mass to reach the true label—i.e., the model is less confident about the correct class.

Table 14: KS Tests: Calibration vs. Test Conformity Score Distributions (seed=42).

Task	KS	p	Mean _{cal}	Mean _{test}
<i>Catastrophic tasks (test scores > calibration)</i>				
s-shipcond	0.956	$< 10^{-10}$	0.54	0.98
s-payterms	0.748	$< 10^{-10}$	0.65	0.92
s-group	0.676	$< 10^{-10}$	0.98	1.00
<i>Robust tasks</i>				
i-plant	0.741	$< 10^{-10}$	0.73	0.86
s-incoterms	0.633	$< 10^{-10}$	0.75	0.87
s-office	0.994	$< 10^{-10}$	1.00	1.00
i-incoterms	0.559	$< 10^{-10}$	0.78	0.66
i-shippoint	0.505	$< 10^{-10}$	0.71	0.58

All KS tests are two-sided. Non-randomized APS scores (conservative coverage guarantee). For catastrophic tasks, test scores shift rightward (stochastic dominance), consistent with Theorem 1. Two robust tasks (i-incoterms, i-shippoint) show test scores shifted *leftward*—the model becomes more confident during COVID, possibly due to reduced product diversity.

I RAPS COMPARISON

Table 15 compares APS and RAPS coverage drops across all 8 tasks with 10-seed means ($\lambda = 0.01$, $k_{\text{reg}} = 5$).

Table 15: APS vs. RAPS Coverage Drop (10-seed means $\pm$ std). $\lambda = 0.01$, $k_{\text{reg}} = 5$.

Task	Cl	APS Drop	RAPS Drop	Δ	Note
<i>Severe tasks</i>					
s-shipcond	45	60.4 $\pm$ 31.8	67.8 $\pm$ 25.4	+7.4	No improvement
s-group	462	73.5 $\pm$ 33.9	10.4 $\pm$ 1.3	−63.1	RAPS protects
s-payterms	135	79.9 $\pm$ 29.4	35.1 $\pm$ 28.6	−44.8	Partial recovery
<i>Non-severe tasks</i>					
i-plant	35	11.8 $\pm$ 9.0	13.1 $\pm$ 5.8	+1.3	Near-identical
i-shippoint	70	9.3 $\pm$ 27.6	20.5 $\pm$ 21.6	+11.2	RAPS worse
i-incoterms	13	10.8 $\pm$ 9.1	10.8 $\pm$ 9.1	0.0	Identical
s-incoterms	13	6.7 $\pm$ 6.8	6.7 $\pm$ 6.8	0.0	Identical
s-office	25	0.04 $\pm$ 0.03	0.04 $\pm$ 0.03	0.0	Identical

All values are 10-seed means $\pm$ std. APS drops differ from 50-seed Table 1 due to smaller seed range (10 vs. 50) and high variance. The key comparison is APS vs. RAPS within the same seed set. RAPS eliminates the catastrophic drop for sales-group by truncating the class accumulation that amplifies score inflation in many-class settings (mean set size: 232/462 classes). Sales-shipcond is unhelped because single-feature dependence places dominant mass on the wrong class *before* the $k_{\text{reg}} = 5$ penalty activates—the exchangeability violation occurs at the top of the APS ordering, not through class accumulation. For low-class tasks (≤ 35 classes), RAPS and APS produce identical or near-identical drops, confirming RAPS regularization has negligible effect when prediction sets are already small. Item-shippoint ($K = 70$) is an exception: RAPS worsens coverage by 11.2 pp (paired Wilcoxon $p = 0.064$, 10 seeds), reflecting that the regularization penalty can disrupt calibration for intermediate-cardinality tasks with high seed variance.

J MODEL CLASS SENSITIVITY

To test whether SHAP concentration is diagnostic across model classes, we replicate the analysis with RandomForestClassifier (500 trees), XGBClassifier, and CatBoostClassifier (ordered boosting), all with seed=42 and the same data pipeline (Table 16).

Results. The diagnostic’s predictive power varies by model class (Table 17). All three gradient-boosted implementations show positive correlation: LightGBM ($\rho = 0.833$, $p = 0.010$), CatBoost ($\rho = 0.667$, $p = 0.071$), and XGBoost ($\rho = 0.548$, $p = 0.16$). RandomForest ($\rho = 0.30$, $p = 0.47$) does not replicate.

Mixed-effects analysis. Since (task, model) pairs share data and temporal split, we fit a linear mixed-effects model with task as a random intercept to properly account for clustering: $\text{Drop}_{ij} = \beta_0 + \beta_1 \cdot \text{Concentration}_{ij} + u_i + \varepsilon_{ij}$, where

Table 16: SHAP Concentration and Coverage Drop Across Model Classes.

Task	LGB	RF	XGB	CB	LGB	RF	XGB	CB
	Concentration (%)				Coverage Drop (%)			
s-shipcond	50.7	57.4	44.6	53.7	71.6	15.8	47.2	1.1
s-group	47.3	22.5	33.3	31.8	71.2	50.0	56.8	14.5
s-payterms	54.2	27.9	30.6	25.7	77.1	13.8	20.4	-5.3
i-shippoint	48.8	52.1	41.0	43.1	18.5	22.7	80.4	61.8
i-plant	23.9	26.8	26.0	29.0	10.6	15.3	9.7	8.5
s-incoterms	23.7	26.8	21.6	28.2	8.5	2.5	3.7	-1.5
i-incoterms	28.9	17.5	18.4	21.7	11.3	2.9	1.1	-0.4
s-office	42.6	24.5	36.8	33.5	0.1	0.0	0.0	-0.0

$u_i \sim \mathcal{N}(0, \sigma_u^2)$ is the task random effect. Across the three gradient-boosted models ($n = 24$, 8 tasks $\times$ 3 models), the concentration effect is $\beta_1 = 1.64$ (95% CI [0.70, 2.58], $p = 0.0006$): each 1% increase in concentration predicts 1.6 pp additional coverage drop, after accounting for task-level clustering. Including model class as a fixed effect yields $\beta_1 = 1.66$ ($p = 0.001$). With all four models including RF ($n = 32$): $\beta_1 = 1.06$ ($p = 0.010$).

Table 17: Concentration–Drop Relationship by Model Class.

Model	Method	Conc.–Drop ρ	p
LightGBM	Boosting (GOSS)	0.833	0.010
CatBoost	Boosting (ordered)	0.667	0.071
XGBoost	Boosting (histogram)	0.548	0.16
RandomForest	Bagging	0.30	0.47
Mixed-effects, 3 boosting ($n = 24$)		$\beta_1 = 1.64$	<0.001
Mixed-effects, all 4 ($n = 32$)		$\beta_1 = 1.06$	0.010

Interpretation. All three gradient-boosted implementations show the concentration–drop relationship, but with different vulnerability profiles. CatBoost’s ordered boosting produces dramatically different task-level vulnerabilities: LGB-catastrophic tasks (s-shipcond: 71.6% LGB drop) show only 1.1% CatBoost drop, while i-shippoint reverses (18.5% LGB $\rightarrow$ 61.8% CatBoost). This confirms each model’s SHAP concentration captures its *own* learned dependence—not a universal data property—and that cross-model transfer is unreliable (LGB conc. $\rightarrow$ CatBoost drop: $\rho = 0.07$). RF’s bagging produces smoother probability surfaces, compressing both concentration range and vulnerability profiles. Specifically, RF’s failure to replicate ($\rho = 0.30$) has two mechanistic causes: (1) *compressed concentration range*—RF concentrations span 17.5–57.4% vs. LGB’s 23.7–54.2%, but the RF range is driven by a single outlier (s-shipcond at 57.4%); excluding it, the RF range is 17.5–26.8%, offering almost no discriminative signal; (2) *smoothed probability surfaces*—bagging averages 500 independent trees, each seeing random feature subsets, which distributes effective dependence more evenly than boosting’s sequential error-correction. The result is that RF rarely develops the concentrated single-feature dependence that drives catastrophic conformal failure in boosting models, and consequently SHAP concentration has limited predictive power for RF vulnerability.

The diagnostic is *model-specific*: it measures how a particular model’s learned dependence structure creates vulnerability. Practitioners should compute concentration for their *deployed* model class rather than transferring results across model families.

Neural network extension. To test beyond tree-based models, we evaluate a 2-layer MLP (128/64 hidden units, ReLU, softmax, permutation importance) on all 8 SALT tasks (10 seeds for sales-level tasks, 3 for item-level). The within-MLP concentration–drop correlation is $\rho = 0.43$ ($p = 0.29$), similar to RF and not significant. However, the model-specificity claim is strikingly confirmed across all 8 tasks: the MLP learns fundamentally different concentration profiles from tree-based models. Key examples: (1) i-shippoint shows MLP $C = 77.2\%$, drop = 82.3 pp vs. LGB $C = 48.8\%$, drop = 18.5 pp—the MLP concentrates nearly all importance on a single feature and fails catastrophically, while LGB distributes importance and maintains coverage; (2) s-incoterms shows MLP $C = 30.3\%$, drop = 42.3 pp vs. LGB $C = 23.7\%$, drop = 8.5 pp—the MLP’s greater concentration predicts its greater failure; (3) s-office shows MLP $C \approx 0\%$, drop = 0.1 pp—perfect robustness prediction. The non-significant within-MLP ρ is driven by two tasks (s-group: $C = 27.5\%$, drop = 78.4 pp; s-payterms: $C = 28.0\%$, drop = 60.6 pp) where the MLP distributes importance evenly yet still fails catastrophically—suggesting the

MLP's failure mechanism involves global sensitivity to the shift rather than concentrated single-feature dependence. This confirms that SHAP concentration is diagnostic specifically for the concentrated-dependence failure mode that characterizes gradient-boosted models, not a universal predictor of all failure modes.